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MILITARY SPECIFICATION

CONNECTORS, TORPEDO TUBE CABLE, MK 47 MOD 0 AND MK 48 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification establishes the requirements for the procurement of Torpedo Tube Cable Connectors Mk 47 Mod 0 and Mk 48 Mod 0 in accordance with NAVORD LD 162874 and/or LD 162875.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-B-117	Bags, Sleeves and Tubing, Interior Packaging
MIL-S-901	Shock Tests, H. I. (High Impact); Shipboard Machinery, Equipment and Systems; Requirements for
MIL-T-18404	Torpedoes, Environmental Requirements, General Specification for

FSC 1355

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-type Base
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

Naval Ordnance Systems Command

LD 162872	Cable, Missile Control, Mk 8 Mod 0
LD 162873	Cable, Missile Control, Mk 9 Mod 0
LD 162874	Connector, Torpedo Tube Cable, Mk 47 Mod 0
LD 162875	Connector, Torpedo Tube Cable, Mk 48 Mod 0
NAVORD Dwg 757981	Container

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The torpedo tube cable connector mates with applicable SUBROC umbilical cables on one side and the missile control cables (NAVORD LD 162872 or LD 162873) on the other side. The connector, which is cylindrical in shape, has 56 hermetically sealed double ended pin contacts. Molded insert assemblies are mated to each side of the double-ended pin contacts and held in place with retaining rings. These molded inserts are removable for cleaning or replacement.

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3.2 General

3.2.1 NAVORD drawings. In addition to the requirements specified herein, the cable connectors shall conform to NAVORD LD 162874 and LD 162875 and all drawings listed thereon.

3.2.2 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Classification of requirements. The requirements for the cable connectors are classified herein as follows:

Requirement	Paragraph
Preproduction	3.4
Materials	3.5
Design and construction	3.6
Mechanical requirements	3.7
Electrical requirements	3.8
Environmental requirements	3.9

3.4 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample (see 4.2.1) of torpedo tube cable connectors is required. The sample shall be manufactured using the methods and procedures proposed for the production lot and will be tested as specified in 4.3.1 for the purpose of determining, prior to starting production, that the manufacturer's production process will produce connectors that meet the requirements of this specification.

3.5 Materials. All materials used in the construction of the connectors shall be as specified herein and as defined on NAVORD LD 162874 and LD 162875.

3.5.1 Material control objective quality evidence for K-Monel components (see 4.6). Objective evidence of chemical analysis and physical properties of material per heat from the mill or per lot from an independent laboratory, traceable from the serial number on the final product, shall be maintained in support of specification compliant material. Chemicals and physicals shall be the result of quantitative laboratory analysis. At least one sample from an identifiable heat or individual piece of bar stock is required for analysis. Resulting analysis shall

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be traceable to and from markings affixed to the material. Material identity, traceable to objective evidence, shall be maintained throughout productive process.

3.5.1.1 Records of heat treatments of K-Monel parts shall be maintained. The heat treatment process and serial number of each part which has been heat-treated shall be included on these records.

3.6 Design and construction. The design and construction of the cable connectors shall be as defined on NAVORD LD 162874 and LD 162875.

3.7 Mechanical requirements. The cable connector shall successfully meet the following mechanical requirement:

3.7.1 Mating or unmating force. The tube door plug of each mating cable and the cable connector shall fully engage or disengage with an axial force of not more than 60 pounds, when tested as in 4.5.2.

3.8 Electrical requirements. The cable connector shall successfully meet the following electrical requirements:

3.8.1 Dielectric strength. The cable connector shall withstand a voltage potential of 1000 volts ac when tested in accordance with 4.5.3.

3.8.2 Insulation resistance. The insulation resistance of the cable connector assembly shall bc greater than 70 megohms when measured with a potential of 500-750 volts dc in accordance with 4.5.3.2.

3.8.3 Continuity. The cable connector shall indicate continuity at each contact when tested in accordance with 4.5.3.3.

3.8.4 Contact, conductor resistance. The maximum resistance of each circuit shall be as shown in Table I when the connector is subjected to the test of 4.5.3.4.

TABLE I

Conductor AWG Size	Max DC Resistance
16	0.4 ohm
12	0.3 ohm

3.9 Environmental requirements. The cable connector assembly shall be subjected to the following environmental conditions and shall meet the requirements as specified herein:

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3.9.1 Hydrostatic pressure. The cable connector shall be capable of withstanding an externally applied pressure of 3000 psi when tested in accordance with 4.5.4.1. During the test, the connector shall show no evidence of water leakage. After being dried with an air blast, the connector shall meet the requirements of 3.8.2 and 3.8.3. This test shall be performed both before and after other tests of 3.9.

3.9.2 Vibration. The cable connector shall successfully withstand a vibration test in accordance with MIL-STD-167, Type I, and 4.5.4.2 of this specification. During this test, there shall be no circuit discontinuities exceeding 100 microseconds in duration. At the conclusion of the test, the unit shall meet the requirements of 3.8.2 and 3.8.3.

3.9.3 Shock. The connector assembly, mounted to simulate service installation, shall be subjected to the shock tests of MIL-S-901 for light weight equipment in accordance with 4.5.4.3. Following this test, the assembly shall meet the requirements of 3.8.2 and 3.8.3.

3.9.4 Salt water immersion. The cable connector, with mating cables attached, shall successfully withstand a salt water immersion test in accordance with 4.5.4.4. Following this test, the assembly shall meet the requirements of 3.8.2 and 3.8.3.

3.9.5 Low temperature (operating). The cable connector shall meet the requirements of 3.7.1, 3.8.2, and 3.8.3 during the low temperature (operating) test defined in 4.5.4.5.

3.9.6 High temperature (operating). The cable connector shall meet the requirements of 3.7.1, 3.8.2, and 3.8.3 during the high temperature (operating) test defined in 4.5.4.6.

3.9.7 Temperature and humidity (operating). The cable connector shall be subjected to the operating temperature and humidity of 4.5.4.7, after which it shall meet the requirements of 3.7.1, 3.8.2, and 3.8.3.

3.9.8 Temperature and humidity (storage mode). The cable connector shall be subjected to the temperature and humidity test of 4.5.4.8, after which it shall meet the requirements of 3.8.2 and 3.8.3.

3.9.9 Tropical environment. The cable connector shall be subjected to the tropical environment of 4.5.4.9, after which it shall meet the requirements of 3.8.2 and 3.8.3.

3.10 Workmanship. The cable connectors furnished to this specification shall be constructed and finished in a manner to assure full compliance with all requirements of this specification and applicable NAVORD drawings.

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Particular attention shall be paid to neatness, damage to O-ring seals, and freedom from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling

4.2.1 Preproduction sampling. Four Mk 47 Mod 0 or four Mk 48 Mod 0 cable connectors shall be supplied by the contractor (if both are being supplied, two of each is sufficient). The connectors shall be manufactured using the same procedures and methods and at the same plant location proposed by contractor for the execution of the contract. The sample shall be submitted to the procuring activity for purpose of determining compliance with the requirements of the contract or order and the applicable specifications. Further production of the item by the contractor prior to the approval of the procuring activity or completion of inspection and test on the preproduction sample shall be at the contractor's risk. Accepted preproduction samples will become the property of the procuring activity and will be included in the number of connectors called for in the contract or order.

4.2.2 Acceptance testing. Acceptance tests as specified herein shall be conducted on each connector assembly.

4.3 Inspection

4.3.1 Preproduction testing and acceptance testing. Preproduction testing and acceptance testing shall consist of the inspections and tests of 4.4.1 and 4.4.2 on the samples selected under 4.2.1 and 4.2.2 as applicable.

4.3.2 Action in case of failure

4.3.2.1 Preproduction sample. In case of failure in any preproduction sample, the cause shall be determined and corrected. A statement of the

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failure and proposed corrective action shall be submitted to the procuring activity for approval before additional tests are conducted.

4.3.2.2 Acceptance tests. In case of failure in acceptance tests, the cause shall be determined and a statement of the failure and proposed corrective action shall be submitted to the procuring activity for consideration and approval. No subsequent testing shall be conducted without the approval of the procuring activity.

4.4 Tests

4.4.1 Preproduction tests. The preproduction test requirements for the cable connectors are shown in Table II. When specified in the contract or purchase order, certified inspection data covering the results of preproduction inspections shall be forwarded to the procuring activity.

TABLE II. PREPRODUCTION TESTS

Examination or Test	Requirement Paragraph	Method Paragraph
Acceptance Tests	4.4.2	4.4.2
Dielectric Strength	3.8.1	4.5.3.1
Hydrostatic Pressure	3.9.1	4.5.4.1
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3
Vibration	3.9.2	4.5.4.2
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3
Shock	3.9.3	4.5.4.3
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3
Salt Water Immersion	3.9.4	4.5.4.4
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3
Low Temperature (Operating)	3.9.5	4.5.4.5
Mating and Unmating Force	3.7.1	4.5.2
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3

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TABLE II. PREPRODUCTION TESTS (Cont'd)

Examination or Test	Requirement Paragraph	Method Paragraph
High Temperature (Operating)	3.9.6	4.5.4.6
Mating and Unmating Force	3.7.1	4.5.2
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3
Temperature and Humidity (Operating)	3.9.7	4.5.4.7
Mating and Unmating Force	3.7.1	4.5.2
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3
Temperature and Humidity (Storage Mode)	3.9.8	4.5.4.8
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3
Tropical Environment	3.9.9	4.5.4.9
Insulation Resistance	3.8.2	4.5.3.2
Continuity	3.8.3	4.5.3.3

4.4.2 Acceptance tests. The manufacturer shall maintain a 100 percent inspection procedure to assure that each connector meets the requirements of Table III.

TABLE III. INDIVIDUAL TESTS

Examination or Test	Requirement Paragraph	Method Paragraph
Examination of Product	3.2, 3.5, 3.6, 3.10	4.5.1
Insulation Resistance	3.8.2	4.5.3.2
Hydrostatic Pressure	3.9.1	4.5.4.1
Connector Mating and Unmating Force	3.7.1	4.5.2
Contact, Conductor Resistance	3.8.4	4.5.3.4

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4.4.3 Test conditions. Unless otherwise specified, tests and examinations required by this specification shall be conducted under any combination of conditions within the ranges below. Any specified condition shall not affect the other two ambient ranges.

(a) Temperature: 20 degrees C to 30 degrees C.

(b) Relative humidity: Room ambient to 95 percent.

(c) Barometric pressure: 24 to 31 inches of mercury.

4.5 Test methods

4.5.1 Examination of product. The cable connector and its components shall be examined to insure conformance to all requirements of this specification and the applicable drawings. Assurance shall be made that no changes have been made in the connector other than those granted for preproduction approval. The connector shall be examined to assure compliance with the following requirements:

(a) 3.2.1 NAVORD drawings.

(b) 3.5 Materials.

(c) 3.6 Design and construction.

(d) 3.10 Workmanship.

4.5.2 Connector mating and unmating force. The connector assembly shall be mounted in a movable holding device capable of measuring force in both a push and pull direction. Each mating cable assembly tube door plug or equal shall be engaged and disengaged to the mating connector by hand as in service use. The force required to engage and disengage the units shall be not greater than 60 pounds. See Figure 1.

4.5.3 Electrical tests

4.5.3.1 Dielectric strength. A voltage of 1000 volts ac shall be applied between two points, one point being each circuit in turn, the other point being all the remaining circuits and the shell tied together to ground. The potential shall be applied for 1 minute. There shall be no indication of arcing during the test.

4.5.3.2 Insulation resistance. A 500-750 volt dc potential shall be applied between two points, one being each circuit in turn, the other being all remaining circuits and the shell connected together to ground. The

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potential shall be applied for 2 minutes maximum or until a stable reading is attained. The instrumentation shall be capable of detecting a leakage current of 10 microamperes or less. The cable connector shall meet the requirements of 3.8.2.

4.5.3.3 Continuity. Each circuit shall be probed with a 6-12 volt dc buzzer, or equivalent, for continuity. There shall be no evidence of discontinuity, to meet the requirements of 3.8.3.

4.5.3.4 Contact, conductor resistance. A missile control cable assembly, NAVORD LD 162872 or LD 162873, shall be mated to the cable connector and the resistance measurement made from each female socket of the connector to the corresponding male pin in the in-line connector (see Figure 2). The ohmmeter or Kelvin Bridge shall be capable of measuring a resistance of less than 0.1 ohm with an accuracy of at least plus or minus 0.02 ohm. The measured resistance of each circuit shall not exceed the values specified in 3.8.4.

4.5.4 Environmental tests

4.5.4.1 Hydrostatic pressure. Remove the snap rings, the insert holder, and the insert assembly from the connector and mount the connector in the hydrostatic pressure tank as shown in Figure 3. The bleeder hole shall be suitably sealed. The side normally connected to the SUBROC umbilical shall be subjected to the hydrostatic pressure. Raise the pressure to 3000 psi as rapidly as practical and hold for 15 minutes. At the end of 15 minutes, the pressure shall be completely released and then raised to 50 psi for a period of 1 minute. At the conclusion of the test, the connector shall be thoroughly cleaned with distilled water and dried with an air blast. After reassembly, the connector shall meet the requirements of 4.5.3.2 and 4.5.3.3.

4.5.4.2 Vibration. Vibration tests shall be performed with the connector in the mated condition. The mated cable assembly and cable connector, with mating connectors, shall be mounted to a hydrostatic pressure tank which simulates a tube door, and this fixture in turn shall be mounted to a sinusoidal vibration system as illustrated in Figure 4. The vibration fixture shall be sufficiently rigid to prevent resonance frequencies greater than 1.5 to 1. All circuits of the mating connectors shall be wired in series. The input control accelerometer shall be placed on the vibration fixture adjacent to the cable connector mounting point. Two monitoring accelerometers shall be mounted to the cable assembly, one on the in-line connector end and one on the tube door end.

4.5.4.2.1 Vibration schedule and levels. Vibration shall be conducted in three mutually perpendicular directions (see Figure 4) in accordance with the following schedule. All tests in one direction shall be completed before proceeding with tests in another direction.

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NOTE: All references to amplitude of vibration refer to maximum displacement of sinusoidal motion from the position of rest, or one-half the total excursion.

(a) Resonance search. The sample shall be vibrated at frequencies from 5 Hertz (Hz) (or lowest obtainable but not higher than 10) to 33 Hz at a table vibratory amplitude of 0.010 ± 0.002 inch and from 34 Hz to 50 Hz at a table vibratory amplitude of 0.003 + 0.00 - 0.00 inch. The change in frequency shall be made in 1-Hz steps and maintained at each step for 15 ± 5 seconds. The frequencies at which the greatest resonance or discontinuities in excess of 100 microseconds occur shall be noted.

(b) Vibration levels. The sample shall be vibrated in accordance with Table IV. The dwell time at each integral frequency shall be 5 minutes.

TABLE IV

Frequency Range (Hz)	Table Amplitude (Inch) (See Note Above)
4 to 15	0.030 ± 0.006
16 to 25	0.020 ± 0.004
26 to 33	0.010 ± 0.002
34 to 40	0.005 ± 0.001
41 to 50	0.003 + 0.000 - 0.001

(c) Resonance dwell. The sample shall be vibrated for 2 hours at the frequency at which the greatest resonance or discontinuities were noted. If no discontinuity or resonance occurred during the search, this test shall be performed at 50 Hz. During the resonance search and dwell portions of the test, hydrostatic pressure shall be applied for 5 minutes. The pressure shall be equal to normal depth firing conditions.

4.5.4.2.2 The above schedule shall be completed in each of the three mutually perpendicular axes. At the conclusion of the vibration tests, the test specimens shall be subjected to the tests of 4.5.3.2 (Insulation resistance) and 4.5.3.3 (Continuity). Immediately following the above tests, the hydrostatic pressure test of 4.5.4.1 shall be performed.

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4.5.4.3 Shock. The cable connector shall be mated to the Missile Control Cable, NAVORD LD 162872 or LD 162873, and securely mounted to a shock fixture simulating service conditions as shown in Figure 4. All circuits shall be wired in series for the purpose of monitoring any discontinuities. The test specimens shall be subjected to the shock test defined in MIL-S-901 for light-weight equipment. Three shocks shall be applied parallel to each of the three principal axes shown in Figure 4. In each axis, the hammer shall be dropped from 1, 3, and 5 feet. A total of nine shocks shall be performed. The series circuit shall be monitored, during shock, for discontinuity, using a relay chatter indicator or equivalent. Discontinuities in excess of 100 microseconds shall be considered evidence of failure.

4.5.4.3.1 At the conclusion of the shock test, the hydrostatic pressure test of 4.5.4.1 shall be performed.

4.5.4.4 Salt water immersion. The connector, mated with SUBROC umbilical and missile control cables, shall be subjected to a 30-day immersion test. The connector shall be attached to a fixture so that all of the connector is immersed in sea water. The connector shall be immersed for 16 hours and removed for a period of 8 hours. This sequence shall be repeated 30 times. No fresh water rinse will be permitted during the 30 cycles. At the conclusion of this test the connectors shall be subjected to the tests of 4.5.3.2 and 4.5.3.3.

4.5.4.5 Low temperature (operating). The cable connector shall be mated to the Missile Control Cable Assembly, NAVORD LD 162872 or LD 162873, and mounted in the test chamber. The chamber temperature shall be lowered to plus 28 degrees F plus or minus 5 degrees F and maintained for a period of 48 hours. At the end of this time, and while still at the test temperature, the test specimen shall be subjected to the tests of 4.5.3.2, 4.5.3.3, and 4.5.2.

4.5.4.6 High temperature (operating). The cable connector shall be mated to the Missile Control Cable Assembly, NAVORD LD 162872 or LD 162873, and mounted in the test chamber. The chamber temperature shall be raised to 130 degrees F plus or minus 5 degrees F and maintained for a period of 48 hours. At the end of this time, and while still at the test temperature, the test specimen shall be subjected to the tests of 4.5.3.2, 4.5.3.3, and 4.5.2.

4.5.4.7 Temperature and humidity (operating). Both SUBROC umbilical and missile control cables shall be mated to each of two cable connectors. The units shall then be placed in a test chamber and subjected to a temperature of plus 122 degrees F for 2 hours. During this 2-hour period the requirements of 4.5.3.2 and 4.5.3.3 shall be met.

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(a) After 2 hours, the receptacles shall be subjected to five 24-hour cycles of 16 hours at plus 122 degrees F and 8 hours at plus 86 degrees F. The relative humidity shall be maintained constant at 95 percent during the five cycles. While at plus 122 degrees F the requirements of 4.5.3.2 and 4.5.3.3 shall be met.

(b) Upon completion of five cycles, the temperature shall be maintained at plus 122 degrees F for 8 hours. During the 8-hour period, the relative humidity shall be 50 percent for the first 2 hours, 95 percent during the next 4 hours, and 40 percent during the final 2 hours. At 30-minute intervals the requirements of 4.5.3.3 shall be met. Upon completion, the requirements of 4.5.2 and 4.5.3.2 shall be met.

NOTE: The transitions between temperatures shall be accomplished within the 8-hour period so that the time at high temperature is approximately 16 hours. Each transition shall not exceed 1½ hours. The relative humidity need not be controlled during the transition periods.

4.5.4.8 Temperature and humidity (storage mode). The cable connector, in the packaged condition, shall be subjected to a 28-day temperature/humidity test. The total test is made up of two 14-day temperature and humidity cycles. The term "chamber temperature," as used in this test, refers to the temperature of the air immediately surrounding the equipment. Chamber temperature may be changed (1) by varying the temperature of a single chamber or (2) by moving the test specimen from one constant temperature chamber to another. The fluctuations from all specified temperatures shall not exceed 10 degrees F. The following test sequence is based on a start at 0800 Monday. Regardless of the actual day and time the test is initiated, there shall be no deviation from the test sequence as prescribed.

(a) At 0800 Monday place the test specimen in a chamber which has been maintained at minus 65 degrees F for at least 2 hours immediately prior to 0800 (humidity not controlled).

(b) At 1600 Monday the chamber temperature shall be changed to plus 160 degrees F and 95 plus 3 minus 0 percent humidity. The chamber temperature must reach plus 160 degrees F and 95 percent humidity not later than 1800.

(c) The test specimen shall be held under these conditions until 0800 Tuesday, at which time the temperature decrease shall begin. The rate of decrease of chamber temperature shall be equal to or greater than

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36 degrees F per hour for at least 2½ hours. Thus, at 1030, the temperature of the air surrounding the test specimens must be plus 70 degrees F or lower. The chamber temperature must reach minus 65 degrees F no later than 1400 and this temperature shall be held until 1600.

(d) At 1600 Tuesday the chamber temperature shall be changed to plus 160 degrees F and 95 plus 3 minus 0 percent humidity as rapidly as practicable (not later than 1800) and held until 0800 Wednesday.

(e) On Wednesday, Thursday, and Friday the operations performed on Tuesday shall be repeated.

(f) The chamber temperature shall be raised to plus 160 degrees F and 95 percent humidity on Friday evening and shall be so maintained until 0800 Monday.

(g) At 0800 Monday the sequence of operations described above for Tuesday of the first week shall be followed and shall be repeated daily until Friday of the second week.

(h) On Friday, the chamber temperature shall be reduced to minus 65 degrees F and this temperature shall be maintained until 0800 Monday of the third week. The cycle will be complete on 0800 Monday.

4.5.4.8.1 The above sequence of temperature and humidity conditions will constitute one cycle. Two such cycles shall be applied. The second cycle shall be completed at 0800 Monday of the fifth week. At that time, the test specimen shall be allowed to return to room temperature. Without drying, wiping, or baking, the test specimen shall meet the requirements of 4.5.3.2 and 4.5.3.3.

4.5.4.9 Tropical environment. For this test, the cable connector shall be mated to LD 162872 or LD 162873. The coupled unit shall be mounted in the test chamber wall or suitably protected so as to prevent exposing the in-line connector face and the tube side of the cable connector.

(a) The test specimen shall be placed in a mold chamber in which a temperature of 86 degrees F plus or minus 5 degrees F and a relative humidity of 90 percent plus or minus 5 percent is maintained. The test specimen, while so housed, shall be sprayed with a suspension of mixed spores. The test specimen shall be maintained in this condition for a period of 28 days. At the end of this period, the test specimen shall be allowed to return to room temperature, and it shall be examined for evidence of being nutrient to fungi. If the test specimen shows susceptibility to fungi, colored photographs shall be taken.

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(b) At least five groups of fungi shall be used in the test specimen inoculation. Five such groups are listed in MIL-T-18404 and one type of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a ph value between 5.8 and 7.2 at 77 degrees F) shall be prepared by sterilization in convenient containers (approximately 100 milliliters each). Approximately 10 milliliters of the solution shall then be introduced directly into a stock culture and shaken vigorously in order that a well sporulated suspension will result without disturbing the agar.

(c) This process shall be repeated for each type fungus. The separate spore suspension from the various types of fungi shall then be mixed together in an atomizer to provide a composite suspension. Actively growing cultures in ripe fruiting condition shall be used for each inoculation and under no condition will the inoculum be kept for more than an 8-hour period.

(d) The cable connector shall show no evidence of being nutrient to fungi and shall meet the requirements of 4.5.3.2 and 4.5.3.3.

4.6 Material control objective quality evidence

4.6.1 Each single cable connector shell shall be accompanied by material control objective evidence that the material meets all the requirements of the applicable specifications. The following information shall be a part of the objective evidence.

(a) Assembly number.

(b) Contract number.

(c) Contractor's identity.

(d) Serial number.

(e) Material specification and heat treatments with a statement to the effect that objective evidence is on file in support of material identification and heat treatment, traceable from the serial number (see 3.5.1).

(f) Brinell hardness (three readings) with acceptable range of readings.

(g) Direct reading of O-ring surface diameter with specified standard.

(h) That "32" finishes are in accordance with the specification and are free of imperfections.

(i) Signature or symbol of contractor's inspection.

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(j) Signature or symbol of Defense Contract Administration Services representative.

4.6.2 The contractor shall pack one copy of the objective quality evidence with the item and retain a duplicate copy on file. The cognizant inspection service is authorized to withhold shipment until the required objective quality evidence is in accordance with this specification. Failure to provide acceptable objective quality evidence shall be cause for rejection of the material.

4.6.3 Material control objective quality evidence in support of product quality shall be maintained by the contractor until completion of the contract and then turned over to the designated Government representative.

4.7 Packaging, packing, and marking. Packaging, packing, and marking shall be examined for compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The connectors shall be protected with a plastic dust cap and be bagged in a transparent bag conforming to characteristics equal to or better than a 0.003-inch wall thickness of Type-1, Class-E, Style-2 of MIL-B-117. Loose component parts shall be protected from damage and may be enclosed in similar plastic bags. The connector shall be packed in 3/4-inch thick cellulosic material conforming to PPP-C-843, Type-II, Class-B and packed in a 8 x 8 x 6 fiberboard container in accordance with Drawing 757981.

5.1.2 Level B. Not applicable.

5.1.3 Level C. The connectors shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The connectors, packaged as described in 5.1.1.1, shall be packed, 12 each, in a wooden box in accordance with PPP-B-621, Class-2 and may be shipped individually in small lots.

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5.2.1.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 95.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The connectors packaged in accordance with 5.1.3 shall be packed to afford protection from damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special markings. None, unless otherwise specified.

5.3.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Torpedo Tube Cable Connectors Mk 47 Mod 0 and Mk 48 Mod 0 are intended for use aboard submarines in transmitting fire control electrical signals through the torpedo tube door by interconnecting SUBROC umbilical cables and Missile Control Cables Mk 8 Mod 0 or Mk 9 Mod 0.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type connectors required (LD 162874 and/or LD 162875).

(c) Assigned activity for preproduction inspection.

(d) Applicable levels of packaging (see 5.1).

(e) Applicable levels of packing (including palletizing) (see 5.2).

6.2.1 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.4.1	Certified inspection data	-
4.6	Material Control Objective Quality Evidence	-

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DIDs (Data Item Descriptions/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Custodian:
Navy - OS

Preparing Activity
Navy - OS
Project No. 1355-N339

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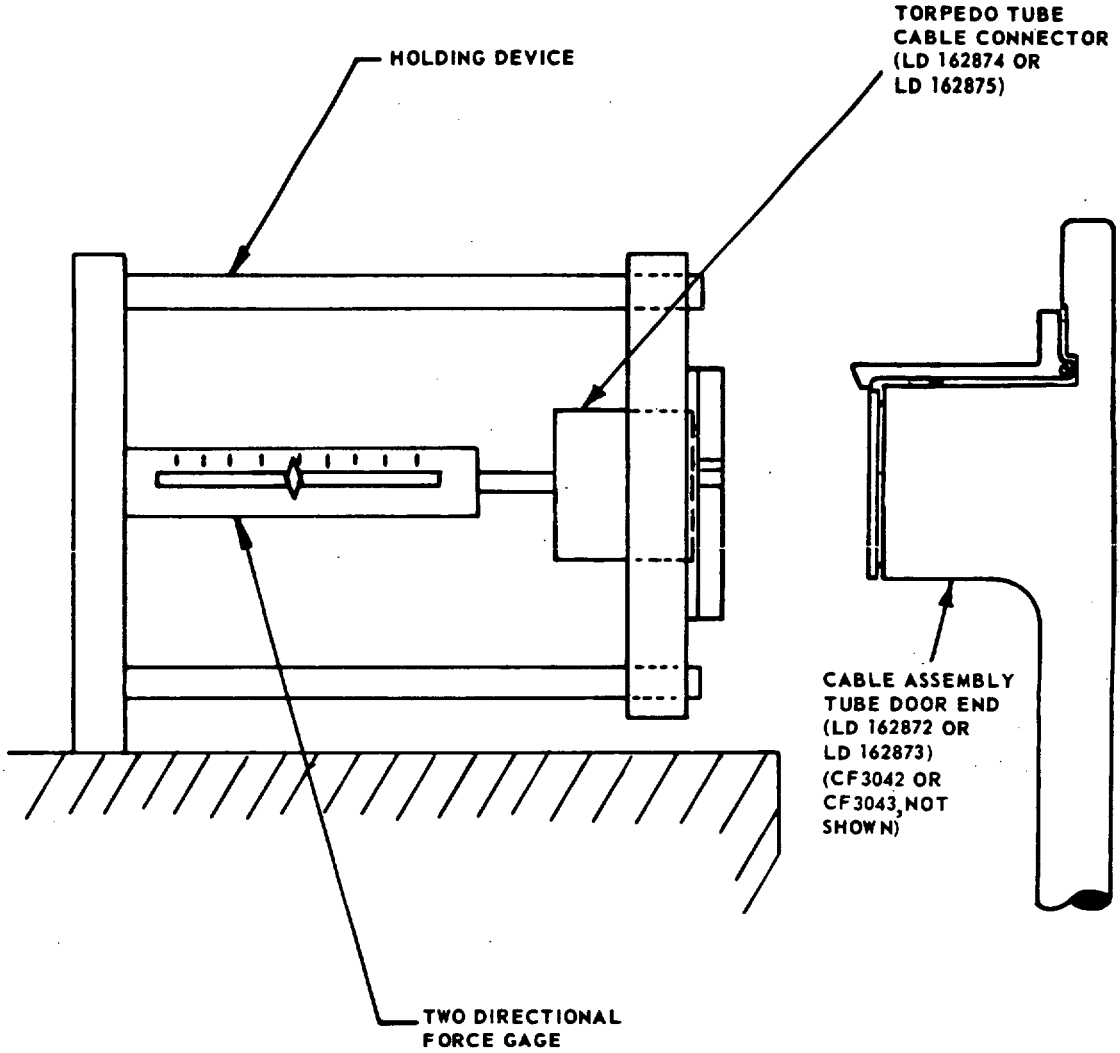


FIGURE 1. TEST SETUP – MATING AND UNMATING FORCE

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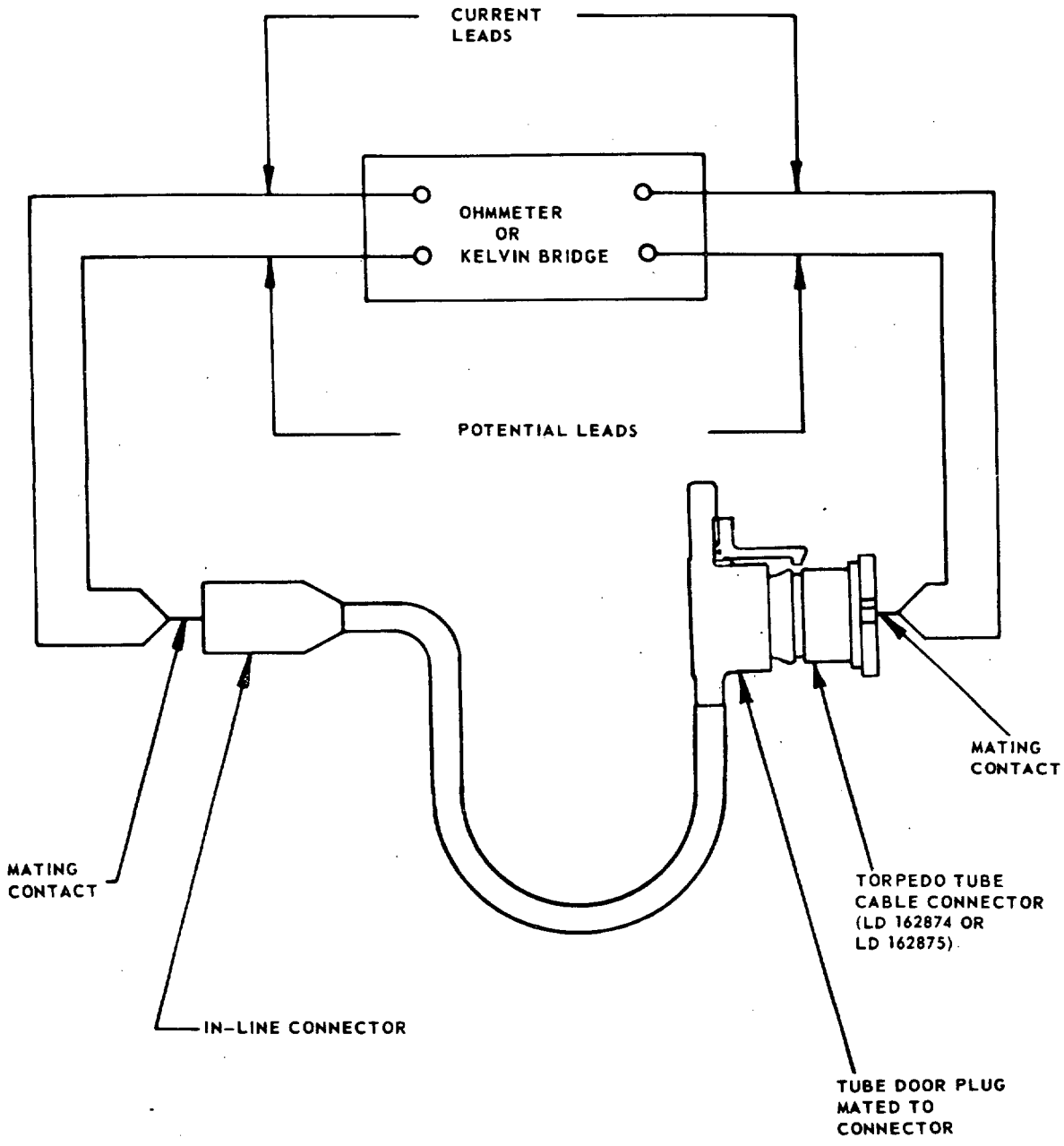


FIGURE 2. TEST SETUP – CIRCUIT RESISTANCE

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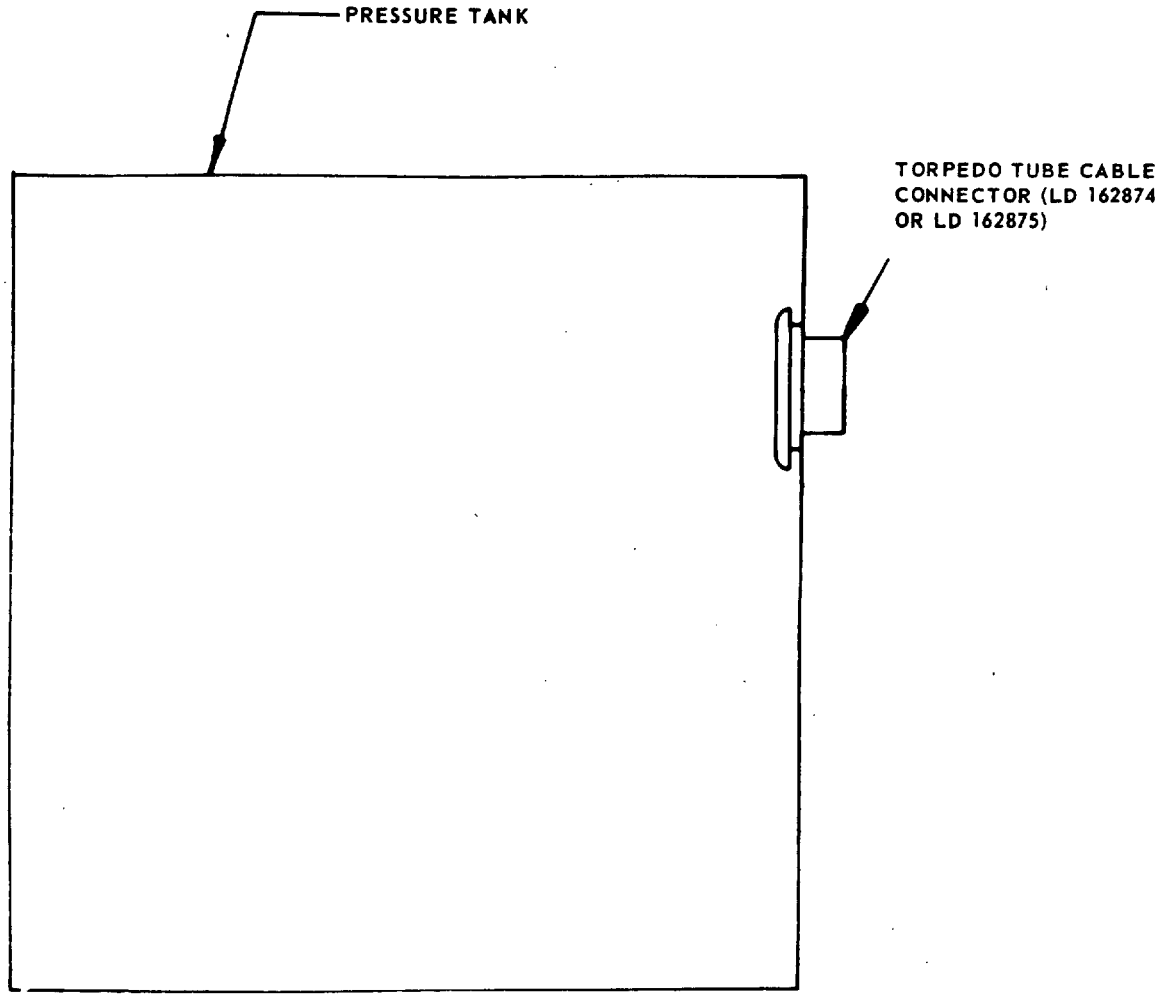


FIGURE 3. TEST SETUP – HYDROSTATIC PRESSURE

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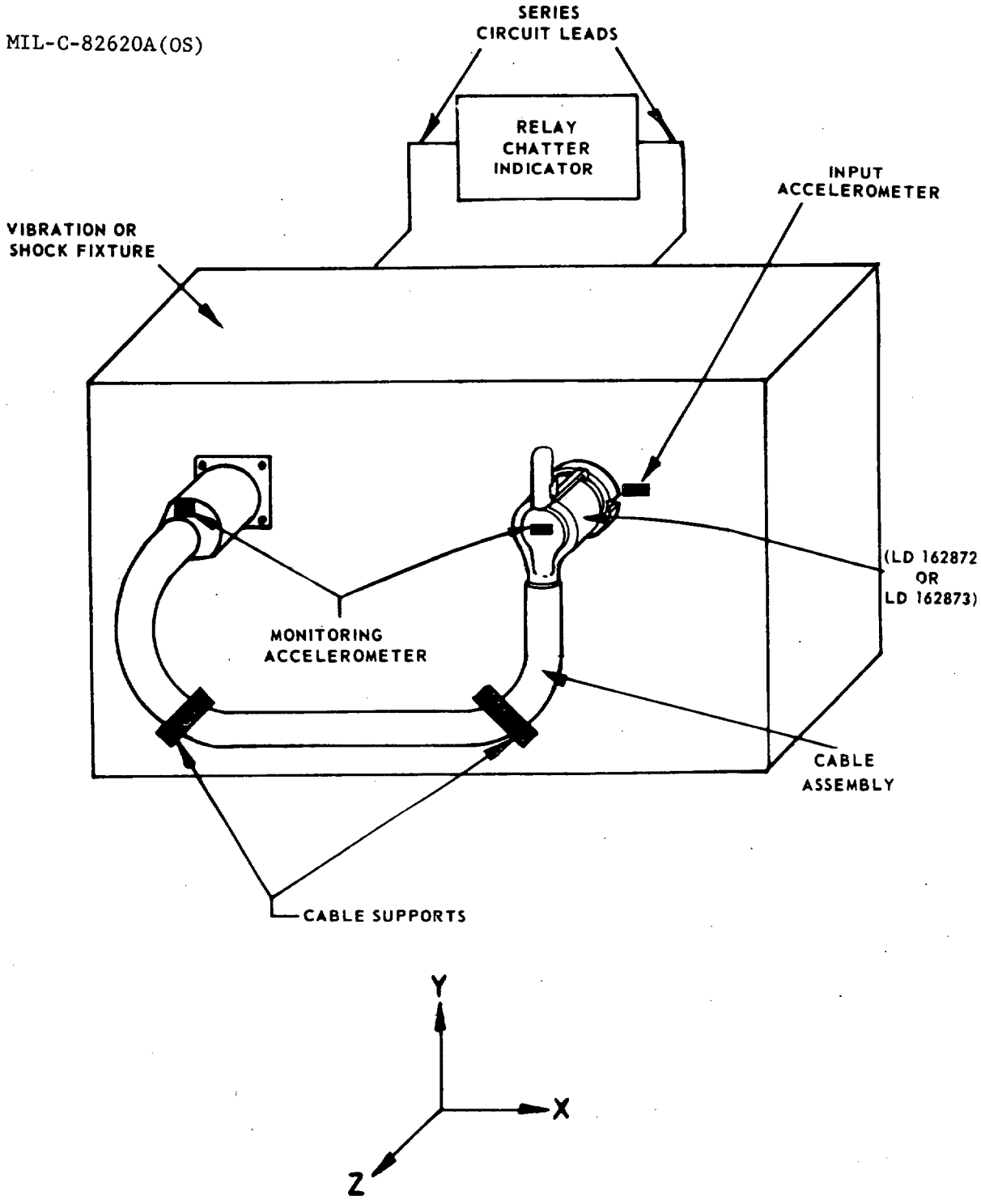


FIGURE 4. TEST SETUP – VIBRATION AND SHOCK

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-C-82620A(OS) Connectors, Torpedo Tube Cable, MK 47 MOD 0 and MK 48 MOD 0

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N 0102-014-1801 C-25254

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Naval Ordnance Station
Standardization Division (502)
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